Pimpri Chinchwad Education Trust's

PIMPRI CHINCHWAD COLLEGE OF ENGINEERING

SECTOR NO. 26, PRADHIKARAN, NIGDI, PUNE 411044

(An Autonomous Institute Approved by AICTE and Affiliated to SPPU, Pune)

Curriculum Structure and Syllabus

of

First Year B. Tech. Computer Science and Engineering (AI & ML)

(Regulations 2023)

Effective from Academic Year 2024-25

Institute Vision

To be one of the top 100 Engineering Institutes of India in coming five years by offering exemplarily Ethical, Sustainable and Value Added Quality Education through a matching ecosystem for building successful careers.

Institute Mission

1. Serving the needs of the society at large through establishment of a state-of-art Engineering Institute.
2. Imparting right Attitude, Skills, Knowledge for self-sustenance through Quality Education.
3. Creating globally competent and Sensible engineers, researchers and entrepreneurs with an ability to think and act independently in demanding situations.

EOMS Policy

"We at PCCOE are committed to offer exemplarily Ethical, Sustainable and Value Added Quality Education to satisfy the applicable requirements, needs and expectations of the Students and Stakeholders.

We shall strive for technical development of students by creating globally competent and sensible engineers, researchers and entrepreneurs through Quality Education.

We are committed for Institute's social responsibilities and managing Intellectual property.

We shall achieve this by establishing and strengthening state-of-the-art Engineering Institute through continual improvement in effective implementation of Educational Organizations Management Systems (EOMS)."

Course Approval Summary

Board of Studies - Department of Computer Science and Engineering (AI & ML)

Sr. No.	Name of the Course	Course Code	Page number	Signature and stamp of BoS chairman
1	Discrete Mathematics	BCS21ES01	18	Chairman, Board of Studies Computer Science & Engineering (AI & ML) PCET's Pimpri Chinchwad College of Engineering Sector 26, Pradhikaran, Nigdi, Pune - 411044.
2	Computer Organization & Operating Systems	BCS21ES02	20	
3	Computer Organization & Operating Systems Lab	BCS21ES03	22	
4	C Programming	BCS21VS01	24	
5	Data Science	BCS22ES04	38	
6	Data Science Laboratory	BCS22ES05	40	
7	Software Engineering	BCS22PC01	43	
8	Object Oriented Programming using C++	BCS22VS02	45	

Board of Studies – Applied Science and Humanities

Sr. No.	Name of the Course	Course Code	Page number	Signature and stamp of BoS chairman
1	Linear Algebra &Univariate Calculus	BSH21BS01	11	Chairman BoS, Applied Sciences & Humanities PCET's, Pimpri Chinchwad College of Engineering Sector No. 26, Pradhikaran, Nigdi, Pune-44
2	Linear Algebra &Univariate Calculus Laboratory	BSH21BS02	13	
3	Engineering Physics	BSH21BS03	14	
4	Engineering Physics Laboratory	BSH21BS04	16	
5	Indian Knowledge System	BSH21IK01	26	
6	Life Skill I	BSH21CC01	27	
7	Multivariate Calculus	BSH22BS07	30	
8	Multivariate Calculus Laboratory	BSH22BS08	32	
9	Engineering Chemistry	BSH22BS05	34	
10	Engineering Chemistry Laboratory	BSH22BS06	36	
11	AEC(Eng/Ger/Jap /Business story telling)	BSH22AE01/ 02/03/04	49/51/53/ 55	
12	Life Skill II	BSH22CC02	58	

Chairman
Academic Council
PCET's, Pimpri Chinchwad College of Engineering
Sector No. 26, Pradhikaran, Nigdi, Pune-44

Approved by Academic Council:
Chairman, Academic Council
Pimpri Chinchwad College of Engineering

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CURRICULUM FRAMEWORK
(2023 Regulations)

LIST OF ABBREVIATIONS

Sr. No.	Abbreviation	Type of Course
1	BSC	Basic Science Course
2	ESC	Engineering Science Course
3	PCC	Program Core Course
4	PEC	Program Elective Course
5	MDM	Multidisciplinary Minor
6	OEC	Open Elective
7	VSEC	Vocational and Skill Enhancement Course
8	AEC	Ability Enhancement Course
9	EEM	Entrepreneurship/Economics/Management Course
10	IKS	Indian Knowledge System
11	VEC	Value Education Course
12	ELC	Experiential Learning Courses
13	CC/LLC	Co-curricular courses / Liberal Learning Courses

COURSE WISE CREDIT DISTRIBUTION

Sr. No.	Type of Course	No. of Courses	Total Credits	
			NO.	%
1	Basic Science Course	8	14	35
2	Engineering Science Course	5	12	30
3	Programme Core Course	1	2	5
4	Vocational and Skill Enhancement Course	2	4	10
5	Ability Enhancement Course	1	2	5
6	Indian Knowledge System	1	2	5
7	Co-Curricular Courses	2	4	10
	Total	**20**	**40**	**100**

SEMESTER-WISE COURSE DISTRIBUTION

Course Distribution: Semester Wise										
Sr. No.	Type of Course	No. of Courses / Semester								Total
		1	2	3	4	5	6	7	8	
1.	Basic Science Course	4	4							8
2.	Engineering Science Course	3	2							5
3.	Programme Core Course	-	1							1
4.	Vocational and Skill Enhancement Course	1	1							2
5.	Ability Enhancement Course	-	1							1
6.	Indian Knowledge System	1	-							1
7.	Co-Curricular Courses	1	1							2
Total		**10**	**10**							**20**

SEMESTER-WISE CREDIT DISTRIBUTION

Credit Distribution: Semester Wise										
Sr. No.	Type of Course	No. of Credits / Semester								Total
		1	2	3	4	5	6	7	8	
1.	Basic Science Course	7	7							14
2.	Engineering Science Course	7	5							12
3.	Programme Core Course	-	2							2
4.	Vocational and Skill Enhancement Course	2	2							4
5.	Ability Enhancement Course	-	2							2
6.	Indian Knowledge System	2	-							2
7.	Co-Curricular Courses	2	2							4
Total		**20**	**20**							**40**

Curriculum Structure

First Year B. Tech.

CSE AIML

CURRICULUM STRUCTURE

First Year B.Tech. CSE (AI & ML) Semester – I

First Year B. Tech CSE (AI & ML) (2023 Regulations) (With effect from Academic Year 2024-25)																
Semester I																
Course Code	Course Name	Credit Scheme				Teaching Scheme (Hours/Week)			Evaluation Scheme and Marks							
		L	P	T	Total	L	P	T	FA		SA	TW	PR	OR	Total
									FA1	FA2					
BSH21BS01	Linear Algebra & Univariate Calculus	2	-	-	2	2	-	-	10	10	30	-	-	-	50
BSH21BS02	Linear Algebra & Univariate Calculus Laboratory	-	1	-	1	-	2	-	-	-	-	25	25	-	50
BSH21BS03	Engineering Physics	2	-	-	2	2	-	-	10	10	30	-	-	-	50
BSH21BS04	Engineering Physics Laboratory	-	2	-	2	-	4	-				50	50	-	100
BCS21ES01	Discrete Mathematics	3	-	-	3	3	-	-	20	20	60	-	-	-	100
BCS21ES02	Computer Organization & Operating Systems	3	-	-	3	3	-	-	20	20	60	-	-	-	100
BCS21ES03	Computer Organization & Operating Systems Lab		1	-	1	-	2	-	-	-	-	50	-	-	50
BCS21VS01	C Programming	-	2	-	2	-	4	-	-	-	-	50	-	50	100
BSH21IK01	Indian Knowledge System	2	-	-	2	2	-	-	25	25	-	-	-	-	50
BSH21CC01	Life Skill I	-	2	-	2	-	4	-	-	-	-	100	-	-	100
Total		12	08	-	20	12	16	-	85	85	180	275	75	50	750

L-Lecture, P-Practical, T-Tutorial, FA–Formative Assessment, SA-Summative Assessment, TW-Term Work, OR-Oral, PR-Practical

***Exit Policy: Available as a separate document**

CURRICULUM STRUCTURE

First Year B.Tech. CSE (AI & ML) Semester – II

First Year B. Tech CSE (AI & ML) (2023 Regulations) (With effect from Academic Year 2024-25)															
Semester II															
Course Code	Course Name	Credit Scheme				Teaching Scheme (Hours/Week)			Evaluation Scheme and Marks						
		L	P	T	Total	L	P	T	FA		SA	TW	PR	OR	Total
									FA1	FA2					
BSH22BS07	Multivariate Calculus	2	-	-	2	2	-	-	10	10	30				50
BSH22BS08	Multivariate Calculus Laboratory	-	1	-	1	-	2	-	-	-	-	25	25	-	50
BSH22BS05	Engineering Chemistry	2	-	-	2	2	-	-	10	10	30	-	-	-	50
BSH22BS06	Engineering Chemistry Laboratory	-	2	-	2	-	4	-	-	-	-	50	50	-	100
BCS22ES04	Data Science	3	-	-	3	3	-	-	20	20	60	-	-	-	100
BCS22ES05	Data Science Laboratory	-	2	-	2	-	4	-	-	-	-	50	50	-	100
BCS22PC01	Software Engineering	2	-	-	2	2	-	-	10	10	30	-	-	-	50
BCS22VS02	Object Oriented Programming using C++	-	2	-	2	-	4	-	-	-	-	50	-	50	100
BSH22AE 01/02/03/04	AEC (Eng/Ger/Jap /Business story telling)	1	1	-	2	1	2	-	10	10	30	-	-	-	50
BSH22CC02	Life Skill II	-	2	-	2	-	4	-	-	-	-	100	-	-	100
Total		10	10	-	20	10	20	-	60	60	180	275	125	50	750

L-Lecture, P-Practical, T-Tutorial, FA–Formative Assessment, SA-Summative Assessment, TW-Term Work, OR-Oral, PR-Practical

***Exit Policy: Available as a separate document**

Course Syllabus

First Year B. Tech.

CSE AI & ML

Semester I

Program:	B. Tech. CSE (AI & ML)					Semester: I	
Course:	Linear Algebra & Univariate Calculus					Code:	BSH21BS01
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge of

1. Elementary Mathematics
2. Elementary Calculus

Course Objectives:

This course aims at enabling students,

1. To strengthen the concept of univariate calculus and mathematical modeling of physical systems using ordinary differential equations
2. To get acquainted with advanced techniques for solving problems related to calculus and ordinary differential equation

Course Outcomes:

After learning the course, the students should be able to:

1. Apply the concept of rank for the solution of the system of equations, linear dependence/independence of vectors and finding Eigen values and Eigen vectors
2. Evaluate the limits of indeterminate forms and representation of a function in an infinite series using successive differentiation, Taylor's and Maclaurin's theorems
3. Formulate the mathematical models related to orthogonal trajectories, electrical circuits and one-dimensional heat flow and solve using ordinary differential equations
4. Solve higher order linear differential equations and Apply it to evaluate the current for electrical circuits

Detailed Syllabus

Unit	Description	Duration [Hrs]
I	**Matrices:** Rank, System of linear equations with applications in electrical circuits, Linear dependence and independence, Linear transformations, Eigenvalues, Eigen vectors.	8
II	**Differential Calculus:** Indeterminate Forms, Taylor's series, Maclaurin's series, Successive differentiation and Leibnitz theorem.	7
III	**Ordinary Differential Equations:** Exact differential equations, Differential equations reducible to Exact form. **Applications:** Orthogonal trajectories, Kirchoff's law of electrical circuits (L-R and R-C circuits), One-dimensional conduction of heat (steady state).	8
IV	**Linear Differential Equations:** Linear differential equation of nth order with constant coefficients, General method, Shortcut methods, Method of variation of parameters, Application of linear differential equations in engineering viz. mass spring system, electrical circuits etc.	7
	Total	**30**

Text Books: Include international Author

1. Higher Engineering Mathematics by B.V. Ramana , 34e, Tata McGraw-Hill.
2. Advanced Engineering Mathematics, by Peter V. O'Neil, 7e, Thomson Learning.
3. Linear Algebra & Univariate Calculus by Team Mathematics, PCCoE, Pune, 1e, Techknowledge Publication

Reference Books:

1. Advanced Engineering Mathematics by Erwin Kreyszig, 9e, Wiley Eastern Ltd.
2. Advanced Engineering Mathematics by S.R.K. Iyengar, Rajendra K. Jain, 4e, Alpha Science International, Ltd.
3. Advanced Engineering Mathematics by M. D. Greenberg, , 2e, Pearson Education.
4. Higher Engineering Mathematics by B. S. Grewal, 43e, Khanna Publication, Delhi

e-sources:

NPTEL Course lectures links:

https://www.youtube.com/watch?v=4QFsiXfgbzM&list=PLbRMhDVUMngeVrxtbBz-n8HvP8KAWBpI5

Program:	B. Tech. CSE (AI & ML)					Semester: I	
Course:	Linear Algebra & Univariate Calculus Laboratory					Code:	BSH21BS02
Credits	Teaching Scheme (Hrs. /Week)			Evaluation Scheme and Marks			
	Theory	Practical	Tutorial	TW	OR	PR	Total
1	-	2	-	25	-	25	50

Prior knowledge:

1. Elementary Mathematics.
2. Elementary Calculus

Course Objectives:

This course aims at enabling students,

1. Understanding MATLAB for mathematical computations, visualization, and data analysis.
2. Ability to apply MATLAB to solve engineering problems related to matrices, ordinary differential equations & differential calculus.

Course Outcomes:

After learning the course, the students will be able to:

1. Solve problems related to matrics, differential calculus, and ordinary differential equations.
2. Develop simple MATLAB program for limits, Taylor and Maclaurin Series, Successive differentiation, and solution of ordinary and Linear differential equations.

Detailed Syllabus

Expt. No.	Suggested List of Experiments
1	Introduction to MATLAB & Matrix operation using MATLAB
2	Properties of Matrices-Rank, Determinant & Solving Linear Systems of Equations
3	Rank & Solution of Linear Systems of Equations using MATLAB
4	Eigen Values and Eigen Vectors
5	Eigen Values & Eigen Vectors using MATLAB
6	Solving ordinary differential equations.
7	Programming Assignment I.
8	Indeterminate Forms
9	Evaluation of limits using MATLAB
10	Taylor's series, Maclaurin's series, and successive derivatives of the function.
11	Expansion of function using MATLAB
12	Solution of linear differential equations.
13	Circuit analysis using linear differential equations.
14	Plotting the solution of linear differential equations using MATLAB
15	Programming Assignment II.

References:

1. Introduction to MATLAB for Engineers and Scientists by Sandeep Nagar, Springer.
2. INTRODUCTION TO MATLAB FOR ENGINEERING STUDENTS by David Houcque, version 1.2, Northwestern University.
3. An Introduction to Differential Equations using MATLAB by Rizwan Butt, Alpha Science International Ltd.

Program:	B. Tech. CSE (AI & ML)					Semester: I	
Course:	Engineering Physics					Code:	BSH21BS03
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge of:

1. Atom, molecule & nuclei,
2. Current, electricity & magnetism,
3. Electromagnetic Induction

is essential.

Course Objectives:

This course aims at enabling students,

1. To build strong conceptual understanding of Semiconductor Physics & Quantum Physics
2. To explore advances in Physics with introduction of Nanotechnology & Superconductivity
2. To provide consciousness about the importance of Physics principles in various engineering applications

Course Outcomes: After learning the course, the students should be able to:

1. Apply basics of semiconductor physics to explain the behavior of charge carriers inside a semiconductor
2. Distinguish wave behavior of a matter particle for the manipulation of the processes at quantum scale
3. Interpret properties of superconductors & their applications in advanced technologies
4. Summarize properties, preparation methods of nanomaterials & explore their applications in various engineering fields

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Semiconductor Physics** Band Theory of solids, Electrical conductivity of conductors & semiconductors (qualitative), Hall effect (with derivation), Fermi Dirac probability distribution function, Fermi level, p-n junction, solar cell I-V characteristics, Basics of sensors, types of sensors-Active and Passive sensors, Applications of sensors.	8
II	**Quantum Mechanics** Limitations of classical physics, need of quantum mechanics, wave particle duality of radiation & matter, De Broglie hypothesis, De Broglie wavelength in terms of kinetic & potential energy, concept of wave packet, phase and group velocity, properties of matter waves, Heisenberg's uncertainty principle, wave function & probability interpretation, well behaved wave function, Schrodinger's time independent wave equation, applications of independent wave equation to the problem of (i) particle in rigid box, (ii) particle in a non-rigid box(qualitative),Tunneling effect, scanning tunneling microscope (STM)	7
III	**Magnetism and Superconductivity** Magnetism: Classification of magnetic materials, temperature dependent magnetic transitions (Curie and Neel temperature), magnetic hysteresis loop, magneto-resistance, giant magneto-resistance (GMR), application of magnetic materials in magneto caloric effect, adiabatic demagnetization. **Superconductivity:** Introduction, critical temperature, properties of superconductors-zero electrical resistance, persistent current, Meissner effect, critical magnetic field, BCS theory, type I and II superconductors, low Tc and high Tc superconductors, Josephson effect, DC-SQUID-construction, working and applications, applications - superconducting magnets, maglev trains	7

IV	**Introduction to Nanoscience** Introduction, surface to volume ratio, quantum confinement, properties of nanomaterials- optical, electrical, mechanical, magnetic; methods of preparation of nanomaterials- bottom-up and top-down approaches, physical methods- high energy ball milling, physical vapor deposition; chemical method - colloidal route for synthesis of gold nanoparticle , aerogels- properties and applications, applications of nanomaterials in medical, energy, automobile, space, defense; introduction to quantum computing.	8
	Total	**30**

Text Books:

1. A textbook of Engineering Physics-Dr. M.N. Avadhanulu, Dr. P.G. Kshirsagar- Revised edition 2015, S. Chand & Company Pvt. Ltd.
2. Engineering Physics-R.K. Gaur, S. L Gupta, -Eighth revised edition 2012, Dhanpatrai Publications (P) Ltd.
3. Nanotechnology -Principles & Practices - Sulabha K. Kulkarni -Third edition -Capital Publishing Company.

Reference Books:

1. Introduction to Quantum Mechanics. - David J. Griffiths, Darrell F. Schroeter, Third edition, Cambridge University Press.
2. Introduction to solid states Physics - Charles Kittel, Eighth Edition, Wiley India Pvt Ltd.
3. Nano: The Essentials. -T. Pradeep, First edition 2007, McGraw Hill Education.
4. Heat and Thermodynamics, Anandamoy Manna, Pearson Publishers 2011

e-sources:

1. NPTEL Quantum Mechanics course: https://nptel.ac.in/courses/122106034
2. NPTEL Fundamentals of semiconductor devices: https://nptel.ac.in/courses/108108122

Program:	B. Tech. CSE (AI & ML)						Semester: I
Course:	Engineering Physics Laboratory					Code:	BSH21BS04
Credits	Teaching Scheme (Hrs. /Week)			Evaluation Scheme and Marks			
	Theory	Practical	Tutorial	TW	OR	PR	Total
2	-	4	-	50	-	50	100

Course Objectives:

1. To provide better understanding of concepts, principles of Physics by giving hands on experience
2. To develop an insight in scientific experimental methodologies.

Course Outcomes:

1. To demonstrate concepts of optics by performing experiments using optical instruments
2. To formulate and solve Engineering Physics problems by applying experimental method.

Guidelines: Group A: Student will perform any eight experiments. Based on this student will be having practical exam of 50 marks

Group B: Student will perform any seven experiments.

Detailed Syllabus

Expt. No.	Group A: Any Eight Experiments
1	To determine the radius of curvature of Plano-convex lens using Newton's rings.
2	To determine unknown wavelength by using plane diffraction grating.
3	To verify Malus Law of polarization of light.
4	To determine refractive indices and identification of types of crystal using double refraction.
5	An experiment based on laser I
6	To determine compressibility of given liquid using Ultrasonic Interferometer
7	To study IV characteristics of solar cell and determine fill factor.
8	To determine band gap of given semiconductor.
9	To determine electrical resistivity of given semiconductor using four probe method
10	To determine Plank's constant "h"
Expt. No.	**Group B: Any Seven Experiments**
1	Testing of properties of semiconductor materials using Hall Effect
2	Testing of magnetic susceptibility of different magnetic material using Quinke's method
3	Testing of sound absorption coefficient of different material
4	Study of solar cell using series and parallel combinations with intensity variation
5	Data analysis using Origin/Mat lab and its interpretation
6	Design and assemble Solar based appliances
7	Laser and fiber optics-based experiment

8	Sensor/Ardino based experiment

Reference Books:

1. Lasers & nonlinear Optics-B. B. Laud-Third edition, New Age International (P)Ltd. Publishers.
2. Fundamentals of Optics- Francis A. Jenkins, Harvey E. White, Fourth edition, McGraw Hill Education (India) Pvt. Ltd.
3. Sensors Handbook- Sabrie Soloman, Second edition, Mc Grew Hill Publications,
4. Fundamentals of Physics- Resnick & Halliday (John Wiley &sons)
5. An introduction to Laser's theory and applications – Dr. M. N. Avdhanulu, Dr. P.S. Hemne– Revised edition 2017-S. Chand & Company Pvt. Ltd.
6. Introduction to solid states Physics - Charles Kittel, Eighth Edition, Wiley India Pvt Ltd.

Program:	B. Tech. CSE (AI & ML)			Semester: I			
Course:	Discrete Mathematics			Code:			BCS21ES01
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
3	3	-	-	20	20	60	100

Prior knowledge of Linear Algebra and Univariate Calculus **is essential.**

Course Objectives:

1. To learn the mathematical formulation of problems by using set, function and relations.
2. To learn logic and proof techniques to reason the solutions mathematically.
3. To interpret set theory, graph theory, and algebraic structures.
4. To learn the formal proof techniques in the formulation of problems.
5. To learn probability theory and various distributions.

Course Outcomes:

After learning the course, the students will be able to:

1. Use fundamental discrete structures and perform various set operations.
2. Solve the problems logically using discrete objects like functions and relations.
3. Use propositional logic to formulate the problems mathematically.
4. Use graph and tree techniques in problem-solving.
5. Identify and apply the basic techniques of counting.
6. Apply probability theory and by analyzing the various types of probability distributions.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Mathematical Reasoning and Set theory** **Fundamentals of logic:** Propositions, Truth Tables, Logical Connectivity, Propositional logic and its applications **Set Theory:** Introduction, Set Representation, Types of Sets, Set Operations, Laws of set theory, Introduction to Structured sets: Groups and Rings. **Mathematical Induction:** Introduction, proof technique, Case Study on Verification for the correctness of computer program using principle of mathematical induction	8
II	**Relations and Functions:** **Relation:** Relation Definition, Properties of Binary Relations, Closure of Relations, Warshall's Algorithm, Equivalence Relations and Equivalence Classes, Partitions, Partial Ordering Relations, Hasse Diagrams and Lattices, Chains and Anti-chains. **Function:** Function Definition, Composition of Functions, Injective, Surjective and Bijective Function, Inverse of a Function	7
III	**Graphs and Trees:** **Graph theory:** Basic Terminology, Types of Graphs, Paths and Circuits, Hamiltonian and Euler Paths and Circuits, Isomorphic Graphs, Planar Graph, Dijkstra's Shortest Path Algorithm. **Trees**: Trees, Rooted Trees, Prefix Codes, Huffman Algorithm for Optimal Tree, Spanning Trees, Minimum Spanning Trees, Kruskal's and Prim's Algorithm.	8
IV	**Counting:** Introduction, Basic counting principles, principle of inclusion and exclusion, pigeonhole principle, permutations, combinations, permutations and combinations with repetitions, binomial theorem.	7

V	**Probability:** The axioms of probability, discrete probability, Random variables, discrete random variable, conditional probability, independent events, Bayes' rule, Bernoulli trials, probability mass function, continuous random variable, probability density function.	7
VI	**Probability Distribution** - Cumulative distribution function, properties of cumulative distribution function. Two-dimensional random variables and their distribution functions, Marginal probability function, Independent random variables. Probability distributions: Gaussian distribution, Binomial Distribution, Poisson poisson distribution.	8
Total		**45**

Textbooks:

1. C. L. Liu, "Elements of Discrete Mathematics", Tata McGraw-Hill, 4th Edition, 2017, ISBN 978- 1259006395.
2. Kenneth H. Rosen, "Discrete Mathematics and its Applications", Tata McGraw-Hill, 8th Edition, 2018, ISBN 978-1259676512

Reference Books:

1. Dr. K. D. Joshi, "Foundations of Discrete Mathematics", New Age International Limited Publishers, 2nd Edition, January 2014, ISBN-13: 978-8122435986
2. Nicholas N.N., Nsowah-Nuamah, "Introduction to Probability Theory" Universal Book Stall.

Program:	B. Tech. CSE (AI & ML)					Semester: I	
Course:	Computer Organization and Operating Systems					Code: BCS21ES02	
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
3	3	-	-	20	20	60	100

Prior knowledge of Computer fundamentals
is essential.

Course Objectives:

1. To get acquainted with the basic concept of computer organization & operating systems.
2. To learn and understand data structures used in the design of operating systems.
3. To understand the functions of operating systems.
4. To learn process management and inter-process communication
5. To learn memory management and I/O management.

Course Outcomes:

After learning the course, the students will be able to:

1. Describe the basic concepts of computer organization.
2. Demonstrate the organizational structure of the central processing unit.
3. Comprehend the concepts of I/O subsystems.
4. Comprehend the internal components of operating systems.
5. Illustrate the concept of process synchronization.
6. Compare various CPU scheduling, disk scheduling, and page replacement policies.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Basic Functional Units of Computers**: Functional units, basic operational concepts, Bus structures, ALU and Control unit. **Data Representation:** Signed number representation, fixed and floating-point Representations, addition and subtraction of signed numbers, signed operand multiplication, booth's algorithm.	7
II	**Central Processing Unit Organization:** Introduction, General Register Organization, Instruction formats, Addressing modes, CISC and RISC, Data path in a CPU, Instruction cycle.	7
III	**Input/output subsystem**: I/O peripherals - Input devices, output devices, I/O ports, Serial port, Parallel port, PCI bus, Secondary storage devices. **Memory Subsystem:** Memory hierarchy, Cache memory.	8

IV	**Introduction to OS:** Operating system functions, Different types of O.S **Process Management:** Process states, Process control block, threads, and system calls. **Scheduling algorithms:** FCFS, SJF, RR, and Priority Scheduling Algorithm.	08
V	**Process Synchronization:** Introduction, Need, Inter Process Communication, producer-consumer problem**,** **Deadlock:** Introduction, Deadlock Prevention, and Deadlock Avoidance.	07
VI	**Memory Management:** Introduction, Memory Allocation Strategies, Swapping, Paging, Segmentation, Virtual Memory, Page Replacement, **Disk management:** Introduction, disk scheduling policies : FIFO, SSTF, SCAN.	08
	Total	**45**

Text Books:

1. W. Stallings, “Computer Organization & Architecture: Designing for performance”, Pearson Education/ Prentice Hall of India,10th Edition, 2016, ISBN-10: 0-13-410161-8 | ISBN-13: 978-0-13-410161-3.
2. Silberschatz A., Galvin P., Gagne G., “Operating System Concepts”, John Wiley and Sons, 9th Edition, 978-1-118-06333-0.

Reference Books:

1. Achyut S. Godbole , Atul Kahate; “Operating Systems", McGraw Hil, 3rd Edition, ISBN-10: 0-07-059113-X| ISBN-13: 978-0-07-059113-4.
2. Dhamdhere D., "Systems Programming and Operating Systems", McGraw Hill, ISBN: 0-07-463579-4.
3. Stallings W., "Operating Systems", 6th Edition, Prentice Hall, ISBN: 978-81-317-2528-3.
4. Andrew S.Tanenbaum, " Modern Operating Systems”, Prentice Hall of India Publication; 3rd Edition. ISBN: 978-81-203-3904-0.

e-sources:

1. https://pdos.csail.mit.edu/6.828/2018/xv6/book-rev11.pdf

Program:	B. Tech. CSE (AI & ML)					Semester: I	
Course:	Computer Organization & Operating System Lab					Code :	BCS21ES03
Credits	Teaching Scheme (Hrs. /Week)			Evaluation Scheme and Marks			
	Theory	Practical	Tutorial	TW	OR	PR	Total
1	-	2	-	50	-	-	50

Prior knowledge of :basic computers **is essential.**

Course Objectives:

1. To comprehend the basic functional units & input-output system of computer organization.
2. To learn the implementation of process management and its scheduling.
3. To understand the concepts and implementation of memory management policies.
4. To analyze and apply various page replacement policies used for paging.

Course Outcomes:

After learning the course, the students will be able to:

1. Comprehend functional units and input-output systems of CPU organization.
2. Demonstrate basic Linux commands & system calls to be used for Shell programming.
3. Implement the various process management policies of CPU scheduling
4. Demonstrate the concept of memory management and its allocation policies.
5. Develop page replacement policies used for paging.
6. Implement intercrosses communication by simulating suitable approaches of process synchronization

Guidelines for Students:

1. The laboratory assignments should be submitted by students in the form of a journal.
2. Each assignment write-up should have a title, objectives, outcomes, Theory- Concept in brief, data description, conclusion, and assessor's signature.
3. Program codes with sample output of all performed assignments should be submitted.

Guidelines for Laboratory / Term Work Assessment:

1. Continuous assessment of laboratory work is done based on the overall performance and laboratory performance of the students.
2. Each laboratory assignment assessment should assign grade/marks based on parameters with appropriate weightage.
3. Suggested parameters for overall assessment as well as each laboratory assignment assessment include- timely completion, performance, innovation, efficiency, punctuality, and neatness.

Guidelines for Laboratory Conduction:

1. The suggested list of assignments is given below. Instructors can frame suitable assignments.
2. Perform any 8 assignments from Group A and Group B. Instructors make sure that all topics get covered.

Suggested List of Assignments

GROUP A - Assignments based on Computer Organization & Basics of Operating System

1	a. To study peripherals of a computer and its functions. b. To study and comprehend the CPU organization & input- output systems of computers.

2	To write a C program for the given problem statements to showcase the conversion between various number systems. a. Decimal to Binary and Binary to Decimal b. Decimal to Octal and Octal to Decimal c. Binary to Hexadecimal and Hexadecimal to Binary
3	To study various system calls & demonstrate those with Linux commands: ls, cat, man, cd, touch, cp, mv, rmdir, mkdir, rm, chmod, pwd.
4	Write a Shell program for the given problem statements: a. To find the factorial of a number. b. To check if the given year is leap year or not c. To check if the given number is even or odd.
5	To study and demonstrate the File / directory related system calls / library functions (read, write, open, close, lseek, opendir, readdir, closedir etc.)
GROUP B - Assignments based on Operating System	
6	To write a C program for implementation of any one scheduling algorithms - FCFS, SJF, Priority, Round Robin).
7	To write a C program to implement the Inter – Process Communication (IPC) technique.
8	To Write a C program to implement any memory allocation technique.
9	To write a C program for the implementation of the FIFO or LRU page replacement algorithm.
10	To write a C program for implementation of FIFO or SSTF disk scheduling algorithms.

Text Books:

1. W. Stallings ,"Computer Organization & Architecture: Designing for Performance,", 10th Edition, 2016, Pearson Education/ Prentice Hall of India, ISBN-10: 0-13-410161-8 | ISBN-13: 978-0-13-410161-3
2. Silberschatz A., Galvin P., Gagne G.; "Operating System Concepts", 9th Edition, John Wiley and Sons.

Reference Books:

1. Dhamdhere D., "Systems Programming and Operating Systems", McGraw Hill, ISBN 0 -07 - 463579 – 4
2. Achyut S. Godbole , Atul Kahate; "Operating Systems", 3rd Edition, McGraw Hill 2
3. Stallings W., "Operating Systems", 6th Edition, Prentice Hall, ISBN-978-81-317-2528-3.
4. Andrew S. Tanenbaum; Modern Operating Systems; Prentice Hall of India Publication; 3rd Edition. ISBN: 978-81-203-3904-0

Web References

1. https://pdos.csail.mit.edu/6.828/2018/xv6/book-rev11.pdf
2. https://nptel.ac.in/courses/106106144

Program:	B. Tech. CSE (AI & ML)						Semester: I
Course:	C- Programming						Code: BCS21VS01
Credits	Teaching Scheme (Hrs. /Week)			Evaluation Scheme and Marks			
	Theory	Practical	Tutorial	TW	OR	PR	Total
2	-	4	-	50	50	-	100

Prior knowledge of mathematics and computers **are essential.**

Course Objectives:

1. To get acquainted with basics of C programming, such as data types, variables, and operators.
2. To learn decision controls and iterations using C programming constructs.
3. To impart the concepts like arrays, strings, pointers, functions, user defined data types and file handling.
4. To develop a simple application using the C language.

Course Outcomes:

After learning the course, the students will be able to:

1. Develop a C program using basic elements like data types, variables and operators.
2. Write the conditional blocks and loops in C programs using suitable constructs.
3. Use C programming elements such as array, string, pointers, functions, user defined data types and file handling libraries.
4. Develop an application based on real life examples using 'C' construct.

Guidelines for Students:

1. The laboratory assignments are to be submitted by students in the form of a journal.
2. Each assignment write-up should have a title, objectives, outcomes, Theory- Concept in brief, dataset used, data description, conclusion, and assessor's signature.
3. Program codes with sample output of all performed assignments should be submitted.

Guidelines for Laboratory / Term Work Assessment:

1. Continuous assessment of laboratory work is done based on the overall performance and laboratory performance of the students.
2. Each laboratory assignment assessment should assign grade/marks based on parameters with appropriate weightage.
3. Suggested parameters for overall assessment as well as each laboratory assignment assessment include- timely completion, performance, innovation, efficiency, punctuality, and neatness.

Guidelines for Laboratory Conduction:

1. The suggested list of assignments is given below. Instructors can frame suitable assignments.
2. All assignments are mandatory.
3. In addition to these, instructors are supposed to assign one real life application in the form of a mini-project based on the concepts learned.

Suggested List of Assignments

Assignment No.	Suggested List of Assignments
1	Write a C program that uses relational operators to compare the temperatures of two cities to determine which one is hotter, colder, or if they have the same temperature.

2	Consider you have created a website in which you are accepting details of users where you have to take password from the user. Write a C program to accept password from user with following condition: 1. Minimum characters 6 and maximum are 12. 2. At least one digit and one character. At least one special symbol (@, $, #).
3	Write a C program to 1. Check if the given number is even or odd. 2. Check if the given number is prime or not. 3. Check if the given number is palindrome or not. 4. Print fibonacci series up to n.
4	Write a C program using a while loop to simulate a dice rolling game. In this game, the player rolls a six-sided dice and accumulates points based on the roll until they choose to stop or roll a 1. The goal is to score as many points as possible without rolling a 1.
5	Write a C program to generate a simple calculator that demonstrates the use of decision control structures: switch statement.
6	Write a C program to manage a student's grades using an array. In this program, allow the user to enter the grades of five subjects for a student and then calculate the average grade.
7	Write a C program to find the length of a string, concatenate two strings, copy string using string library functions.
8	Write a C program to swap the values of two variables using pointers. In this scenario, imagine you are a teacher, and you want to swap the positions of two students in a class. You have the student Names and their seat numbers stored in variables, and you want to swap their seats.
9	Write a C program using functions to simulate a simple banking system. Implement functions for depositing, withdrawing, and checking the account balance.
10	Write a C program that uses a structure to model a real-life example of a book record system. Define a structure to represent a book with attributes like title, author, and year of publication. Perform operations like adding a new book, displaying book details, and searching for a book by title.
11	Write a C program to perform file operations like create, open, read, write and close a file.
12	Design a mini project to implement Game, Smart Text editor or various real time applications integrating C programming concepts.

Text Books:

1. Yashavant Kanetkar , “Let Us C”, BPB publications,17th edition,2020, ISBN-10: 9389845688.
2. Brian W Kernighan, Dennis M Ritchie, “C Programming Language”, 2 nd Edition, Pearson,1988, ISBN: 9780133086249.
3. E. Balagurusamy, “Programming in ANSI C”,McGraw Hill, 8 th Edition,2019, ISBN-13: 978-9351343202.

Reference Books:

1. Herbert Schildt, “C: The Complete Reference”, McGraw Hill, 4th Edition, 2000, ISBN-13: 978-0072121247.
2. Maureen Sprankle , “Problem solving and programming concepts”, Pearson,7th Edition, 2011, ISBN-13: 978-0132492645.
3. R. G. Dromey, “How to Solve it by Computer”, Prentice-Hall International,1 st Edition, 1982, ISBN -13: 978-0134340012.

Web References

1. https://onlinecourses.nptel.ac.in/noc19_cs42/preview
2. https://www.w3schools.com/c/

Program	B. Tech. CSE (AI & ML)						Semester: I
Course:	Indian Knowledge System (IKS)						Code:BS H21IK1
Credit	Teaching Scheme (Hrs./Week)			Evaluation Scheme			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	25	25	-	50

Prior Knowledge: Nil

Course Objectives:
This course aims at enabling students,
1. To familiarize with the concepts of Indian Knowledge System
2. To get acquainted with the applications of Indian Knowledge System

Course Outcomes: After learning the course, the students will be able to;
1. Demonstrate elementary knowledge of various ancient Indian Knowledge Streams
2. Identify/Select Indian Knowledge Stream of Interest
3. Evaluate the chosen aspect Indian Knowledge Stream which in turn may be of assistance in the career of the learner

Detailed Syllabus:

Units	Description	Duration (Hrs)
I	**Indian Knowledge System – Philosophy and Culture** **A. Philosophy** Introduction to Sāṃkhya, Vaiśeṣika, Nyaya, Pūrva-Mīmāṃsā, Vedānta, Ashtanga Yoga–Yamas, Niyamas, Āsana, Prāṇāyāma, Pratyāhāra, Dhāraṇā, Dhyāna, Samādhi **B. Culture** Foundational aspects of Sangeeta, Natya Shastra, Shilpashastra	**15**
II	**Indian Knowledge System – Economics, Science and Technology** Ayurveda, Architecture and Planning, Metallurgy and Material Science, Astronomy, Kautalya Arthashatra	**15**
Total		**30**

Text Books:

1. Mahadevan,B., Bhat Vinayak Rajat, Nagendra Pavanan R.N. (2022), "Introduction to Indian Knowledge System: Concepts and Applications",PHI Learning Private Ltd. Delhi.
2. Dharampal (2021), "Indian Science and Technology in the Eighteenth Century",ISBN10:8175310936.

Reference Books:

Kapil Kapoor, Avadhesh Kumar Singh. (2005), "Indian Knowledge Systems", (Vol.1 and Vol.2), ISBN-10:9788124603369.

Program:	B. Tech. CSE (AI & ML)					Semester: I	
Course:	Life Skills I					Code:	BSH21CC01
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	TW	OR	PR	Total
2	-	4	-	100	-	-	100

Prior knowledge of: Nil

Course Objectives:

1. To equip them with essential competencies that complement their academic education, preparing them to excel not only as engineers but also as well-balanced individuals.
2. To develop students' vital life skills that promotes personal growth, resilience, and success in their academic journey and beyond.

Course Outcomes:

After learning the course, the students will be able to:

1. Demonstrate self-awareness and inner harmony conducive to understanding the essence of happiness.
2. Exhibit proficient interpersonal skills in fostering and sustaining healthy relationship with others.
3. Employ diverse strategies for rational decision-making and problem solving.
4. Display enhanced emotional intelligence through the recognition and management of emotions in various contexts.

Course Guidelines:

1. The students are instructed to perform all the activities suggested by the course faculty.
2. The assessment of life skills activities will be done on the basis of students' performance, attitude, behavior and understanding of subject.
3. The students are suggested to attend all the sessions of the life skills course.

Detailed Syllabus

Unit	Description	Duration (Hrs)
I	**Happy You, Happy Life!** Healthy Mind - Music Therapy, Yoga, Meditation, Happiness and Success, Self-Awareness - Know your personality, Develop your Self- Esteem, *Johari* Window, SWOT, Setting goals for yourself (SMART), Healthy Lifestyle - Nutrition, Significance of Physical Activity in Daily routine. ● Activity/Worksheet 1: Music Therapy ● Activity/Worksheet 2: Johari Window ● Activity/Worksheet 3: Physical & Mental Health	15
II	**Building Relationships** People Skills - Networking, Developing Healthy Relationships, Collaboration, Reliability, Respectfulness, Open- Mindedness, Effective Communication in Relationships-My Relationship Web, Relationship Recipe, Active Listening and Conflict Resolution, Embracing Diversity: Respect for Different Perspectives and Cultures. ● Activity/Worksheet 4: Building Relationships ● Activity/Worksheet 5: Building String Relationships- The Key to Success in Business & Life	15
III	**The Reflective Engineer** Critical Thinking - Fact or Fiction, Convergent & Divergent Thinking, Creative Thinking - Imagination, Formulate and Articulate Ideas, Perspective Thinking – Understanding others view Points, Respecting Others Opinions,	15

	Decision Making – Rational, Analytical & Ethical Solutions. ● Activity/Worksheet 6: Creative & Critical Thinking ● Activity/Worksheet 7: Decision Making ● Activity/Worksheet 8: Perspective Thinking	
IV	**You CAN DO IT…** Managing Stress - Good Stress , Bad Stress, Anxiety, Managing time - Planning, Prioritization, Delegation, Productivity and Positivity, Managing Emotions – Self-Regulation, Self-Motivation, Empathy, Assertiveness, Anger Management, Dealing with addictions, Handling Peer Pressure- Types of Peer Pressure: Spoken Peer Pressure, Unspoken Peer Pressure, Direct Peer Pressure, Indirect Peer Pressure, Positive Peer Pressure, Negative Peer Pressure. ● Activity/Worksheet 9: Managing Emotions ● Activity/Worksheet 10: Stress Management & Peer Pressure	15
	Total	**60**

Reference Books:

1. The 7 Habits of Highly Effective Teens" by Sean Covey Publisher: Simon & Schuster, 2017
2. How to Win Friends and Influence People" by Dale Carnegie Publisher: Simon & Schuster. 2020
3. Emotional Intelligence: Why It Can Matter More Than IQ" by Daniel Goleman Publisher: Bantam Books, 2021.
4. Mindset: The New Psychology of Success" by Carol S. Dweck Publisher: Ballantine Books, 2019.
5. The Power of Habit: Why We Do What We Do in Life and Business" by Charles Duhigg Publisher: Random House, 2016

E Sources -

1. Psychology Today (www.psychologytoday.com): Psychology Today publishes articles and insights from psychologists and mental health experts that can be useful for improving life skills and emotional intelligence.
2. Lifehack (www.lifehack.org): Lifehack shares practical tips, techniques, and advice on personal development, productivity, and life skills improvement.
3. Coursera (www.coursera.org): Coursera offers online courses on various life skills topics, often provided by universities and experts, to help individuals develop essential skills

"Knowledge Brings Freedom"
Progress Credibility Confidence
Optimism Excellence
Since 1996

Course Syllabus

First Year B. Tech.

CSE AI & ML

Semester II

Program:	B. Tech. CSE (AI & ML)					Semester: II	
Course :	Multivariate Calculus					Code :	BSH22BS07
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge of

1. Elementary Mathematics
2. Elementary Calculus

is essential.

Course Objectives:

This course aims at enabling students,

3. To strengthen the concepts of multivariable calculus and its application in maxima & minima, error & approximation, area and volume
4. To make students acquainted with advanced Mathematical techniques to represent Fourier series and to evaluate integrals

Course Outcomes: After learning the course, the students should be able to:

1. Evaluate Partial Differentiation and apply the concept of partial differentiation to find Maxima & Minima and Error & Approximation
2. Represent Fourier series for the periodic time domain continuous and discrete function into signal form
3. Compute definite improper integrals like Gamma, Beta function, DUIS
4. Apply multiple integration techniques to find Area and Volume

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Partial Differentiation:** Partial derivatives, Composite function, Chain Rule, variable to be treated as constant, total derivatives. Euler's theorem for homogeneous functions. **Application of Partial derivatives:** Jacobian for explicit function, Errors and Approximations, Maxima and Minima of two variable functions.	8
II	**Fourier Series:** Definition, Dirichlet's conditions, full range Fourier series, Harmonic analysis, and engineering application.	7
III	**Integral Calculus:** Beta and Gamma functions, differentiation under integral sign (DUIS).	7
IV	**Multiple Integral:** Double integration, conversion into polar form, application of double integration to the area, Triple integration, Dirichlet's theorem, application of triple integration to Volume.	8
	Total	**30**

Text Books:

1. Higher Engineering Mathematics by B.V. Ramana (Tata McGraw-Hill)
2. Advanced Engineering Mathematics, 7e, by Peter V. O'Neil (Thomson Learning)

Reference Books:

1. Higher Engineering Mathematics, 22e, by H. K. Das (S. Chand Publication, Delhi).
2. Advanced Engineering Mathematics by Erwin Kreyszig (Wiley Eastern Ltd.
3. Advanced Engineering Mathematics, 4e, by S.R.K. Iyengar, Rajendra K. Jain (Alpha Science International, Ltd)
4. Advanced Engineering Mathematics, 2e, by M. D. Greenberg (Pearson Education)
5. Higher Engineering Mathematics by B. S. Grewal (Khanna Publication, Delhi)

e-sources:

1. NPTEL Multivariable Calculus course https://nptel.ac.in/courses/111107108
2. NPTEL Video for Fourier series http://nptel.iitm.ac.in

Program:	B. Tech. CSE (AI & ML)						Semester: II	
Course:	Multivariate Calculus Laboratory						Code:	BSH22BS08
Credits	Teaching Scheme (Hrs. /Week)			Evaluation Scheme and Marks				
	Theory	Practical	Tutorial	TW	OR	PR	Total	
1	-	2	-	25	-	25	50	

Prior knowledge of

1. Elementary Mathematics
2. Elementary Calculus
3. Basics of MATLAB/Open Source

is essential.

Course Objectives: This course aims at enabling students to solve problems based on concepts

1. Multivariate calculus and its application in maxima & minima, error & area, and volume.
2. Continuous and discrete systems require knowledge of Fourier series and Harmonic analysis.
3. Advanced techniques to evaluate multiple integrals

Course Outcomes: After completion of this course, the students will be able to,

1. Solve problems related to the concepts of multivariate calculus, such as partial differentiation and its application, Fourier series, integral calculus, and the application of multiple integrals.
2. Use MATLAB/Open source software to solve problems such as partial differentiation, maxima-minima, Fourier series, and multiple integrals.

Detailed Syllabus

Expt. No.	Suggested List of Experiments
1	Problems on Partial derivatives, Euler's theorem on homogeneous functions, implicit functions, and variables treated as constant, total derivatives.
2	Partial derivatives of two variable functions by using MATLAB
3	Problems on Jacobians and their applications, errors, and approximations. Maxima and Minima: maxima and minima of functions of two and three variables.
4	Maxima and Minima(global and local) by using MATLAB with visualization
5	Problems on Jacobians and errors and approximations.
6	Absolute and Relative error for two variable functions by MATLAB
7	Assignment on Programming-1
8	Problems on half-range Fourier series, Harmonic analysis, and application to engineering
9	Fourier series and its plots by using MATLAB
10	Problems on Beta and Gamma functions, differentiation under integral sign (DUIS),
11	Problems on Introduction of curve tracing, double integration, change of order of integration, conversion into polar form, Triple integration: with limits and without limits, Dirichlet's theorem
12	Three-dimensional Cartesian, polar, and cylindrical systems by using GeoGebra
13	Evaluation of Double and triple Integration by MATLAB
14	Problems with evaluation of Area and volume. Visualization of Area, Volume by Geogebra
15	Assignment on Programming-2

References:

1. Higher Engineering Mathematics by H. K. Dass, 22e, S. Chand Publication, Delhi.
2. Advanced Engineering Mathematics by S.R.K. Iyengar, Rajendra K. Jain, 4e, Alpha Science International, Ltd.
3. Advanced Engineering Mathematics by Peter V. O'Neil, 7e, Thomson Learning.
4. Advanced Engineering Mathematics by M. D. Greenberg, 2e, Pearson Education.
5. Higher Engineering Mathematics by B. S. Grewal, 43e, Khanna Publication, Delhi
6. Introduction to MATLAB for Engineers and Scientists by Sandeep Nagar, Springer.
7. Introduction to MATLAB for engineering students by David Houcque, version 1.2, Northwestern University.

Program:	B. Tech. CSE (AI & ML)					Semester: II	
Course :	Engineering Chemistry					Code:	BSH22BS05
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge of:

1. Structure of water.
2. Volumetric analysis.
3. Fossil and derived fuels.
4. Corrosion and its effects.
5. Electrochemical series.
6. Classification and properties of polymers

Course Objectives:

This course aims at enabling students,

1. To familiarize students with instrumental methods for qualitative and quantitative analysis and explore the importance of green chemistry.
2. To build consciousness about the recent development in alternative energy sources and batteries
3. To make student acquainted with chemical and electrochemical mechanism of corrosion and corrosion control
4. To lead students to investigate the advancement in engineering materials.

Course Outcomes:

After learning the course, the students should be able to:

1. Analyze the water quality, interpret techniques of water purification and compare green over traditional synthesis of polycarbonate.
2. Recognize the fuel quality and understand the scope of derived alternate fuels
3. Apply the preventive methods of corrosion to real-life problems.
4. Understand the chemical structure and properties of various polymers, nanomaterials and their uses.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Water Technology and Green Chemistry**: a) Hardness of water, its types, units of hardness and hardness calculation. Chemical analysis of water by determination of hardness by EDTA method. Alkalinity of water and its determination. Numerical on EDTA method and alkalinity. Disadvantages of hard water in boilers. Water softening techniques: Permutit and Ion exchange method. Dissolved oxygen (DO), biological oxygen demand (BOD) and Chemical oxygen demand (COD). b) Introduction of Green Chemistry: Definition, goals, principles and green synthesis of Polycarbonate	8

II	**Fuels and combustion** a) Fuels: definition, calorific value and its units. Calorific value (CV), gross calorific value (GCV), net calorific value (NCV). Determination of calorific value - Bomb calorimeter, Boy's colorimeter and numerical. i) Solid fuels: coal, proximate and ultimate analysis of coal, numerical based on analysis of coal. ii) Liquid fuels: composition of petroleum, refining of petroleum. Synthesis, properties, advantages and disadvantages of Power alcohol and Biodiesel. iii) Gaseous fuels: Hydrogen gas as a future fuel, production by steam reforming of methane and by electrolysis of water. Challenges in storage and transportation of H_2 gas. b) Combustion: chemical reactions, calculations on air requirement for combustion.	7
III	**Corrosion and Corrosion control** a) Corrosion: introduction, types of corrosion, mechanism of atmospheric corrosion and wet corrosion. Galvanic series. Factors affecting corrosion: nature of metal and nature of environment. Different types of corrosion: Pitting corrosion, concentration cell corrosion, stress corrosion and soil corrosion. b) Corrosion control: methods of prevention of corrosion - cathodic and anodic protection, metallic coatings and its types - anodic and cathodic coatings. Method to apply metallic coatings - hot dipping, cladding, electroplating and cementation.	7
IV	**Chemistry of Polymers and Novel Carbon Compounds** a) Polymers: definition, classification of polymers on the basis of thermal behaviour, properties of polymers: degree of polymerization, crystallinity, Tg& Tm and factors affecting Tg. Polymerization and its types. Advanced polymeric materials: Structure, properties and applications of liquid crystal polymer – Kevlar, conducting polymers - Polyacetylene, electroluminescent polymer – PPV and biodegradable polymers – PHBV. b) Nanomaterials: definition, types of nanomaterials and properties of nanomaterials. Quantum dots: Types, properties and applications of QDs. Structure properties and applications of Graphene and Carbon Nano Tubes (CNTs).	8
	Total	**30**

Text Books:

1. Engineering Chemistry by S.S. Dara, S.Chand Publications (2010).
2. Engineering Chemistry by B.S. Chauhan, UnivScPress.(2015).
3. A TextBook Of Engineering Chemistry by Shashi Chawla, DhanpatRai& Co. (2015).
4. Nanotechnology: principles and practices by S.K. Kulkarni, Springer (2014).
5. Engineering Chemistry by Jain and Jain, DhanpatRai Publishing Co.(2016).
6. Engineering Chemistry by Wiley India (2012).
7. Engineering Chemistry by O.G. Palanna, McGraw-Hill Education.
8. Introduction to Nanoscience and Nanotechnology by K. K. Chattopadhyay, A. N. Banerjee. PHI Learning (2009).

Reference Books:

1. Hydrogen as a fuel by Ram D. Gupta, C.R.C.Publication (2009).
2. Polymer Science by V.R.Gowariker,, New Age International Publication (2015).
3. Nanotechnology by T. Gregory, Springer Verlog New York (1999).
4. Introduction to Nanotechnology by Charles P. Poole, Frank Owens, John Wiley & Sons (2003)
5. Engineering Chemistry by Wiley India Pvt.Ltd,First edition 2011.

Program:	B. Tech. CSE (AI & ML)					Semester: II	
Course:	Engineering Chemistry Laboratory					Code:	BSH22BS06
Credits	Teaching Scheme (Hrs. /Week)			Evaluation Scheme and Marks			
	Theory	Practical	Tutorial	TW	OR	PR	Total
2	-	4	-	50	-	50	100

Prior knowledge of
1. Theory of acids and bases
2. Molarity, normality and molality
3. Titration method

is essential

Course Objectives:
1. To help students to procure conceptual clarity of Engineering Chemistry through laboratory experiments.
2. To develop experimental skills to acquire insight into societal and environmental issues.

Course Outcomes:
After completion of this course, the students will be able to,
1. Volumetric analysis for determination of quality of water.
2. Apply various instrumental methods for quantitative and qualitative chemical analysis.
3. Demonstrate the skill for synthesis of engineering materials.
4. Learn the chromatographic separation technique and impact of corrosion.
5. Explore mini projects which are relevant to societal and environmental issues, to develop research attitudes.

Guidelines:
1. Under Group A category, students have to perform all experiments from the list given below.
2. Under Group B category, students have to perform five experiments from the list given below and one mini project .

Detailed Syllabus

Expt. No.	Group A: Suggested List of Experiments
1	Determination of total hardness (by EDTA method) and alkalinity of given water sample.
2	To determine the dissociation constant of a weak acid (acetic acid) using a pH meter.
3	Titration of mixture of strong acid with strong base using Conductivity meter and determine strength of acid.
4	To determine the maximum wavelength of absorption of KMnO4, verify Beer's law and find concentration of the unknown sample.
5	Structural elucidation of unknown compounds by applying principles of UV and IR spectroscopy
6	Proximate analysis of Coal.
7	To determine the electrochemical equivalent (ECE) of Cu.
8	To prepare the Phenol formaldehyde resin.
9	Chromatographic separation of ortho- and para nitro-phenol.
10	Study of corrosion of metal in various mediums
Expt. No.	**Group B: Suggested List of Experiments**
1	Safety in the Chemistry Laboratory
2	To determine the chloride ion (Cl-) present in a given water sample by argentometric method/ Determination of residual chlorine in water.
3	To estimate the amount of Fe (II) present in the given solution potentiometrically.

4	To synthesize p-bromoacetanilide from acetanilide as per Green Chemistry.
5	Determination of acid value of oil.
Topics for Mini project: **(Student has to choose one of the topics from list given below but not limiting to)**	
1	Adsorption studies of Methylene blue on bio adsorbents prepared from agricultural waste.
2	Synthesis of nano-materials
3	Determination of active ingredients from medicines / concentration of dyes in commercial beverages using UV
4	Water audit of water samples
5	One-pot synthesis of biologically active compounds
6	Microwave assisted chemical reactions.
7	Soil analysis of agricultural soil samples
8	Adulterants in food materials.
9	Colloidal synthesis of 2-6 or 3-5 semiconductor quantum dots nanoparticles.
10	Detection of presence of carbohydrates, fats and proteins in given foodstuffs.
11	Preparation of biodiesel.

References:

1. Vogels Text book of Qualitative Chemical Analysis by J.Mendham, R,C,Denny, J.D.Barnes, M.J.K.Thomas, 6 e, Pearson Education ltd.
2. Applied Chemistry Theory and Practice by O.P.Virmani and A.K.Narula, 2e, New age International (P) Ltd.

Program:	B. Tech. CSE (AI & ML)					Semester: II	
Course:	Data Science					Code:	BCS22ES04
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
3	3	-	-	20	20	60	100

Prior knowledge of Mathematics **is essential.**

Course Objectives:

1. To solve numerical problems using the concepts of linear algebra and matrices.
2. To demonstrate the use of mathematical techniques in the field of data science.
3. To explain the theory of statistics and its use in statistical inference.
4. To perform the data preprocessing operations.
5. To demonstrate the use of data visualization and data analytics techniques.

Course Outcomes:

After learning the course, the students will be able to:

1. Analyze the need of data preprocessing/visualizations operations and apply those to prepare the data
2. Use linear algebra and matrices for problem-solving.
3. Perform statistical analysis to understand the data representation by computing various statistical measures.
4. Apply various hypothesis tests on sample data and provide inferences.
5. Analyze and solve the regression and classification problems for data analysis.
6. Apply the model evaluation techniques using different performance measures.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Introduction to Data Science:** Data Science Life Cycle, Data Analytics and Types, Key performance indicators of data science projects. **Data Preprocessing:** Need, preprocessing techniques for: handling redundant data, data transformation, replacing or handling missing data, and handling data inconsistency. **Data Visualization:** Introduction, need, Data visualization techniques: line plot, scatter plot, histogram, density plot, box plot, bar plot, pie chart, heatmap, etc.	**8**
II	**Mathematics for data science:** **Linear Algebra:** Introduction to Linear Algebra, Matrices and Vectors in Data Science.Vector – vector space, Column vectors, row vectors, working with vectors. Matrices: types of matrices, Addition of Matrices and their properties, Matrix multiplication and their properties. Linear combination as a matrix-vector product.	8

III	**Mathematics for data science:** **Matrices:** Transpose of Matrices, properties of Transpose. Determinants; Determinants and their properties, Cramer's Rule, Inverses. Introduction Minors and Cofactors in a Determinant, Properties of Determinants, Differentiation of a Determinant, Rank of a Matrix, Systems of linear equations, Row reduction, Echelon form and its Properties, Eigenvalues and Eigenvectors.	7
IV	**Statistics and Inference:** Measures of central tendency, Measures of variability: standard deviation, variance, quartiles, Interquartile range. Skewness and Kurtosis. **Sampling & Inference:** Sampling: Introduction, types of sampling. Hypothesis testing, types of errors, level of significance, test of hypothesis: t-test, z-test, chi-square, ANOVA. **Correlation analysis:** Pearson's correlation coefficient.	8
V	**Regression:** Understanding Linear regression. Lines of regression – numerical problems. Performing a linear Regression: Linear function, Fitting the line, residual errors, coefficient of determination. Multiple linear regression. Understanding Logistic regression. Performing a logistic Regression: Logistic function, fitting the curve, understanding the log-odds, R-Squared. Multivariate logistic Regression.	7
VI	**Data Analytics and Model Evaluation:** Introduction to classification, Naive Bayes classifier. Model Evaluation and Selection: holdout method, random sub-sampling, cross-validation. Model's parameter tuning and optimization. Performance metrics for evaluation of model, confusion matrix, AUC-ROC analysis. Model building and validations for Naïve Bayes classifier and for logistic regression.	7
	Total	**45**

Text Books:

1. Thomas Nield, "Essential Math for Data Science", O'Reilly Media Inc., October 2022, ISBN: 9781098102869.
2. Data Science and Big Data Analytics, EMC education services, Wiley publication, 2015, ISBN: 9781118876138.

Reference Books:

1. Peter Bruce, Andrew Bruce, Peter Gedeck , "Practical Statistics for Data Scientists", O'Reilly Media, 2nd edition, May 2020, ISBN: 9781492072942.
2. Norman Matloff ,"Probability and Statistics for Data Science", CRC Press, 2019, ISBN:9780429687112.

Web references:

1. NPTEL Course on "Python for Data Science" : https://onlinecourses.nptel.ac.in/noc22_cs74/preview
2. NPTEL Course on "Data Science for Engineers" : https://onlinecourses.nptel.ac.in/noc22_cs72/preview

Program:	B. Tech. CSE (AI & ML)					Semester: II	
Course:	Data Science Laboratory					Code:	BCS22ES05
Credits	Teaching Scheme (Hrs. /Week)			Evaluation Scheme and Marks			
	Theory	Practical	Tutorial	TW	OR	PR	Total
2	-	2	-	50	-	50	100

Prior knowledge of Computer Programming & Problem Solving **are essential**

Course Objectives:

1. To write software programs using linear algebra, matrices, and using data structures in Python.
2. To apply mathematical techniques in the field of data science for analysis of data.
3. To use the theory of statistics for data analysis.
4. To learn the data preprocessing and data visualization techniques.
5. To perform regression analysis and classification tasks for various applications.

Course Outcomes:

After learning the course, the students will be able to:

1. Implement data preprocessing techniques to prepare the data for analysis.
2. Implement data visualization techniques to analyze data graphically.
3. Perform statistical analysis to understand the data representation by computing statistical measures.
4. Write a software program to perform hypothesis tests on sample data.
5. Implement the regression techniques for data analysis.
6. Develop a classification model and evaluate its performance.

Guidelines for Students:

The suggested list of assignments is given below. Instructors can frame suitable assignments based on scenarios. Students should perform all assignments compulsorily using Python. As per assignment requirement, an open-source data can be downloaded from the web (Kaggle, github, UCI ML repository, OpenML, Datahub, Data.gov, etc).

Guidelines for Laboratory / Term Work Assessment:

1. Continuous assessment of laboratory work is done based on the overall performance and laboratory performance of the students.
2. Each laboratory assignment assessment should assign grade/marks based on parameters with appropriate weightage.
3. Suggested parameters for overall assessment as well as each laboratory assignment assessment include- timely completion, performance, innovation, efficiency, punctuality, and neatness.

Suggested List of Assignments

Assignment No.	Suggested List of Assignments

1	Write programs using Python language constructs. Suitable 2 assignments will be framed based on topics such as: Data types and data structures in Python– lists, array, dictionary, tuple, etc. Reading data from files. Different types of language constructs: operators, control structures, functions. Python libraries like pandas, NumPy, etc. Suggested list is as follows: a. Write a python program to create a list of random numbers in normal distribution and display the frequency of each value. b. Write a python program to print the odd and even numbers from a given range taken as per user input and display the prime numbers from the first 20 odd numbers. c. Write a python program to read any .csv file as per user-provided input and display its content.
2	Write programs using Python language constructs. Suitable 2 assignments will be framed based on topics such as: Data types and data structures in Python– lists, array, dictionary, tuple, etc. Reading data from files. Different types of language constructs: operators, control structures, functions. Python libraries like pandas, NumPy, etc. Suggested list is as follows: a. To reverse the tuple or unpack the tuple into multiple variables or access a specific value from tuple b. To merge two dictionaries or create a dictionary by extracting only key values from any given dictionary.
3	Write programs using Python language constructs and Python libraries like NumPy to perform linear algebra operations. Suitable any 2 assignments can be framed. The suggested list is as follows: a. To perform addition and subtraction of matrices b. To perform multiplication and division of matrices
4	Write programs using Python language constructs and Python libraries like NumPy to perform linear algebra operations. Suitable any 2 assignments can be framed. The suggested list is as follows: a. To find the transpose of a matrix b. To solve linear systems of equations by implementing matrices and vectors
5	Perform the following data preprocessing operations using Python and pandas on any open-source dataset. a. Data preprocessing operations: check for missing values, noisy data, null values, etc. and handle those. b. Data Integration and Data Normalization: Summarize the types of variables by checking the data types (i.e., character, numeric, integer, factor, and logical) of the variables in the data set. If variables are not in the correct data type, apply proper type conversions. Perform data transformations suitably like normalization, standardization, etc. Turn categorical variables into quantitative variables
6	Create an “Academic performance” dataset of students and perform the following operations using Python. a. Scan all variables for missing values and inconsistencies. If there are missing values and/or inconsistencies, use any of the suitable techniques to deal with them. b. Scan all numeric variables for outliers. If there are outliers, use any of the suitable techniques to deal with them.

7	Create an "Academic performance" dataset of students and perform the following operations using Python. a. Apply data transformations on suitable variables. The purpose of this transformation should be one of the following reasons: to change the scale for better understanding of the variable, to convert a non-linear relation into a linear one, or to decrease the skewness and convert the distribution into a normal distribution.
8	Write a program for Data Visualization using IRIS flower dataset or any other suitable public dataset . a. Create a histogram for each feature in the dataset to illustrate the feature distributions. Plot each histogram. b. Create a boxplot for each feature in the dataset. All of the boxplots should be combined into a single plot. Compare distributions and identify outliers.
9	Write a program for Data Visualization to create any 6 types of graphs. Select the right chart for data visualization as per requirements like visualization required for comparisons, identification of relationships, identification of trends over time, part of a whole, understanding the distribution of data values, etc. Create graphs among types such as Line plot, Scatter plot, bar plot, Density plot, pie chart, bubble plot, heat maps, tree map, correlation matrices, dendrograms, etc. Use Pima Indians Diabetes dataset or time series data or any other suitable public dataset.
10	Write a python program to perform descriptive statistics to compute various statistical measures by using Pandas library. Use IRIS flower dataset or any other suitable public dataset.
11	Write a program to perform hypothesis testing for a suitable dataset. Select appropriate hypothesis testing strategies such as t-test, z test, chi-square test, ANOVA test, etc.
12	Write a program to implement linear or logistics regression using any suitable public dataset OR the following scenario of Bigmart Sales Analysis may be considered: Perform prediction for the sales of a store using linear or logistics regression. Sample test data set is available here https://datahack.analyticsvidhya.com/contest/practice-problem-big-mart-sales-iii/
13	Write a program to perform classification using Naïve Bayes classifier. Calculate the probabilities by implementing the Bayes theorem using Pima Indians Diabetes dataset or any other suitable public dataset. Measure the performance of the classifier for separate test and training data using various metrics.

Text Books:

1. Thomas Nield ," Essential Math for Data Science" - ,Released October 2022 Publisher(s): O'Reilly Media, Inc. ISBN: 9781098102869.

Reference Books:

2. Peter Bruce, Andrew Bruce, Peter Gedeck, "Practical Statistics for Data Scientists"- · 2020 , ISBN:9781492072898, 1492072893 ,Published:10 April 2020 , Publisher:O'Reilly Media.
3. Norman Matloff, "Probability and Statistics for Data Science" - Norman Matloff · 2019, ISBN:9780429687112, 0429687117 , CRC Press, Published:21 June 2019
4. Chirag Shah, "A Hands-On Introduction To Data Science", Cambridge University Press,(2020), ISBN : ISBN 978-1-108-47244-9.
5. Wes McKinney, "Python for Data Analysis", O' Reilly media, ISBN : 978-1-449-31979-3.

Program:	B. Tech. CSE (AI & ML)					Semester: II	
Course:	Software Engineering					Code:	BCS22PC01
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50

Prior knowledge of Computer Programming and Problem Solving is **essential.**

Course Objectives:

1. To learn the principles of Software Engineering
2. To learn process models for software project development
3. To learn the methods of capturing, specifying, and analyzing software requirements.
4. To learn design principles to software project development

Course Outcomes:

After learning the course, the students will be able to:

1. Understand the principles of Software Engineering and apply appropriate process models for specific software project development.
2. Analyze software requirements and apply architectural principles and design methods for software development.
3. Analyze the agile software development methodologies.
4. Understand CASE, CASE tools and techniques, SPI, Reverse Engineering and Web Engineering.

Detailed Syllabus :

Unit	Description	Duration [Hrs]
I	**Introduction and Software Process Models :** Software Engineering, Myths, Software Process, Software development life cycle, Work Products, Importance of Software Engineering, Standard for Software Process, Waterfall Model, Prototyping Model, Iterative Enhancement Model, Spiral Model, RAD model, 4th Generation models, Formal Methods, Agile Development Case Study: Introduction to Safe Home.	8
II	**Software Requirement Specification, Design and Coding:** Requirements Engineering- Problem analysis, Establishing the Groundwork-Eliciting Requirements, Requirements Analysis-Requirements Modeling Strategies.Design Concepts, Modularity, Architectural design, Coupling and Cohesion, Top-down and bottom-up design,Software Design Document, Coding styles and documentation.	7
III	**Agile development:** Agile Process- Extreme Programming in agile development, Pair Programming in agile development, Agile software development process Models: SCRUM, Sprint Cycle, Sprint Cycle Stages, SCRUM master, Kanban Boards and Methodology, Comparison of Agile with Conventional process models.	8

IV	**Computer Aided Software Engineering and Advanced Topics:** Computer Aided Software Engineering (CASE) and its Scope, CASE support in Software Life Cycle, Architecture of CASE Environment, Upper CASE and Lower CASE, Exposure to CASE tools. Software Process Improvement, Component Based Software Engineering, Web Engineering, Reverse Engineering, Software Engineering challenges of Big Data, Mobile Applications	7
	Total	**30**

Textbooks:

1. Roger S Pressman, "Software Engineering – A Practitioner's Approach", Pearson Education, 7th Edition, 2010.
2. Ian Sommerville, "Software Engineering", 9 th edition, 2010.
3. N.S. Gill, Software Engineering,, Khanna Publishing Co., Delhi 2018.
4. Software Engineering (3rd ed.), By K.K Aggarwal & Yogesh Singh, New Age International Publishers, 2007.

Reference Books:

1. Carlo Ghezzi, "Fundamentals of Software Engineering", Prentice Hall India, ISBN 10: 0133056996, 2002.
2. Rajib Mall, "Fundamentals of Software Engineering", Prentice Hall India, ISBN 13: 978-8120348981, 2014.
3. Pankaj Jalote, "An Integrated Approach to Software Engineering", Springer, ISBN 13: 9788173192715, 2010.

Program:	B. Tech. CSE (AI & ML)					Semester: II	
Course:	Object Oriented Programming using C++					Code:	BCS22VS02
Credits	Teaching Scheme (Hrs. /Week)			Evaluation Scheme and Marks			
	Theory	Practical	Tutorial	TW	OR	PR	Total
2	-	4	-	50	50	-	100

Prior Knowledge of : Computer Programming and Problem Solving is essential.

Course Objectives:

1. To develop solutions by using various OOP features for various applications.
2. To strengthen problem-solving ability by using the characteristics of an object-oriented approach.
3. To develop programming skills using Object oriented concepts.

Course Outcomes:

After learning the course, students will be able to:

1. Apply various object-oriented concepts for problem-solving.
2. Apply inheritance and polymorphism to write object-oriented programs.
3. Develop the solutions using generic templates, exception handling, and STL.
4. Write programs to perform file-handling operations.

Guidelines for Laboratory Conduction:

- Instructor is expected to teach Object oriented programming concepts before conducting assignments.
- Assignments should be implemented in C++ language.
- Operating System recommended: - 64-bit Open source Linux or its derivative.
- Programming tools recommended: - G++/GCC, Eclipse,VS Code, QT creator.
- All Assignments are compulsory.

Assignment No.	Suggested List of Assignments (All assignments are compulsory)
1	**Concepts:**variables and constants, Class – Data members, Member Functions, and class as abstract data type, Object Visibility Modes, Array of Objects. **Assignments:** Write a program Illustrating Class Declarations, Definition, and Accessing Class Members. 1. Write a C++ program to implement a class called Circle that has private member variables for radius. Include member functions to calculate the circle's area and circumference.(Class, Object, Member Function and Access specifier) 2. Create a class 'Student' with three data members which are name, age and address. The constructor of the class assigns default values to name as "unknown", age as '0' and address as "not available". It has two functions with the same name, 'setInfo". First function has two parameters for name and age and assigns the same whereas the second function has three parameters, which are assigned to name, age and address respectively. Print the name, age and address of 10 students.(array as object)
2	**Friend Class, Friend Function, Static data Members, Static Function, Inline Function:** Write a Program Using following concepts: a. Friend Class, b. Friend Function, c. Static data Members, d. Static Function, e. Inline Function

	1. Create a class called Box with a variable: width of type double. Inside the class define a constructor and a friend that prints the width value(printWidth). In the main() define a Box instance, set values and call printWidth. (Friend Function) 2. Design a class A and a class B. Make class B a friend of class A. Implement member functions in both classes to demonstrate the use of friend class. 3. Develop a class Counter to count the number of objects created. Use a static data member to keep track of the count. Implement member functions to increment and display the count. 4. Create a class Math with static functions for addition, subtraction, multiplication, and division. Demonstrate the use of static functions to perform mathematical operations without creating an object of the class.
3	**Constructor:** Default constructor, parameterized constructor and copy constructors and Destructor, Order of Constructor and Destructor Call. **Assignments:** 1. Write a program and input two integers in main and pass them to default constructor of the class. Show the result of the additon of two numbers. (Default Constructor) 2. Create a class called Rectangle.It should contain a constructor for width and height, a function that returns the area and a friend fucntion that duplicate the rectangle's dimensions. In the main, creaate a rectangle instance, duplicate it and print out the area. (Parameterized and Copy Constructor) 3. Write a program to illustrate default constructor, parameterized constructor and copy constructors. Program should illustrate the order of execution of constructors and destructors when a new class is derived from more than one base class.
4	**Inheritance and its Types:** Simple, Multilevel, Hierarchical, Multiple and Hybrid**,** Ambiguity in multiple inheritance. **Assignments:** Write C++ programs that illustrate how the following forms of inheritance are supported: a. Single inheritance b. Multiple inheritance c. Multilevel inheritance d. Hierarchical inheritance. 1. Write a program with a mother class and an inherited daughter class.Both of them should have a method void display ()that prints a message (different for mother and daughter).In the main, define a daughter and call the display() method on it. 2. Write a program with a mother class animal. Inside it define a name and an age variables, and set_value() function.Then create two bases variables Zebra and Dolphin which write a message telling the age, the name and giving some extra information (e.g. place of origin).
5	**Polymorphism :** Operator Overloading : Unary and Binary, Virtual Function, Function Overloading **Assignments:** 1. Implement a class Complex, which represents the Complex Number data type. Implement the following operations: a. Constructor (including a default constructor which creates the complex number 0+0i). b. Overloaded operator + to add two complex numbers. c. Overloaded operator * to multiply two complex numbers.

	d. Overloaded << and >> to print and read Complex Numbers.(Operator Overloading) 2. Write a program defining class Distance that stores feet and inches. Overload operator + to add two instances of class Distance and return the result in another instance. Overload comparison operator < which will compare two instances of class Distance and return true or false. Overload += to add two distances and store the result in the one of them. Overload = operator to assign one distance to the other. (Operator Overloading) 3. Write a C++ program to implement a class called Shape with virtual member functions for calculating area and perimeter. Derive classes such as Circle, Rectangle, and Triangle from the Shape class and override virtual functions accordingly. (Virtual Function) 4. Implement a class StringOperations with overloaded functions to concatenate two strings, to concatenate a string with an integer, and to reverse a string. Test these functions with various inputs to demonstrate their versatility.
6	**ADT Using Class, Pure virtual function, Abstract class.** 1. Write a C++ program to implement the matrix ADT using a class. The operations supported by this ADT are: a. Reading a matrix. b. Addition of matrices. c. Printing a matrix. d. Subtraction of matrices. e. Multiplication of matrices.
7	**Generic Programming:** Class Template and Function Template, Class template with Multiple parameters **Assignments:** Write a class template to represent a generic vector. Include member functions to perform the following tasks: a. To create the vector. b. To modify the value of a given element. c. To multiply the vector by a scalar value. d. To display the vector in the form (10, 20, 30)
8	**Exception Handling:** Fundamentals multiple catching nested try statements uncaught exceptions throw and rethrow 1. Write a C++ program that takes two numbers as input from the user and performs division. Implement exception handling to catch division by zero errors and prompt the user to enter valid inputs. 2. Write a C++ to handle the exception if account balance is less than minimum balance. 3. Write a program that prompts the user to enter their age. Implement exception handling to ensure that the input is a valid integer and handle any ValueError exceptions that may occur.
9.	**File Handling:** Write a program to maintain an employee database in a binary file with employee information such as empId, name, age, department, post and salary. Write function for adding new records, displaying all records, searching for a particular employee, updating employee salary and post.
10.	**Standard Template Library: Stack, Queue, Map** 1. Write a C++ program using STL to perform various stack and queue operations. 2. Write C++ program using STL map for managing Person Records (Name, birth date, telephone no). Perform operations – add, display, search, delete, and update.

	3. Write C++ program using STL to add binary numbers (assume one bit as one number); use STL stack.

Text Books:

1. Richard F. Gilberg& Behrouz A. Forouzan, “Data Structures, Pseudo code Approach with C”, Cengage Learning India

   Edition, 2nd Edition, 2007, ISBN 10: 8131503143 / ISBN 13: 9788131503140.
2. Herbert Schildt, “C++: The Complete Reference”, McGraw Hill Education, 4th Edition, July 2017, ISBN-10 : 007053246X

   (ISBN-13 : 978-0070532465).

Reference Books:

1. Y. Langsam, M. Augenstin and A. Tannenbaum, “Data Structures using C”, Pearson Education Asia, First Edition, 2002, ISBN 978-81-317-0229-1.
2. Bjarne Stroustrup, “The C++ Programming language” , Pearson Education , Third edition, 2008, ISBN 9780201889543.

Program	B. Tech. CSE (AI & ML)					Semester: II	
Course:	AEC-1-English					Code: BSH22AE01	
Credit	Teaching Scheme (Hrs./Week)			Evaluation Scheme			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	1	2	-	10	10	30	50

Prior Knowledge: Basic knowledge of English Language is essential.

Course Objectives: This course aims at enabling students,

1. To develop basic LSRW skills for effective communication.
2. To develop a sense of confidence among students to present themselves at professional as well as societal level.
3. To enhance the language competence.

Course Outcomes: After learning the course, the students will be able to;

1. Understand the role of effective listening skills, grammar and vocabulary in effective communication.
2. Formulate grammatically correct sentences and Enrich their vocabulary
3. Demonstrate reading skills to comprehend various documents
4. Communicate effectively and enhance their phonetic skills.

Detailed Syllabus:

Units	Description	Duration (Hrs)
I	**Listening Skills:** Importance of Listening Skills, Types of Listening: Active / Selective / Passive Listening, Barriers to Listening, Tips to Improve Listening Skills.,	**3**
II	**Writing Skills:** Grammar & Vocabulary: Common Errors in English, Modal Auxiliaries. Processes of Word Formation, Words often Confused Elements of Effective Writing, Writing Styles (Formal & Informal), Paragraph Writing (Descriptive, Technical). Professional Writing: Job Application, Leave Application, Enquiry and Complaint Letter. Features of Technical Writing, Report Writing	**4**
III	**Reading Skills:** Importance of Reading, Scanning, Skimming, Reading between the Lines, Reading Comprehension: Factual / Expository / Informative texts, Case Studies, Reading Research Articles. Lesson:1 The Story of An Hour by Kate Chopin, Lesson: 2 The Classical Student by Anton Chekhov.	**4**
IV	**Speaking Skills:** Basic Sounds-IPA, Word Stress, Intonation, Language Functions (Requesting, Apologizing, Complaining, Complementing, Thanking, etc) Art of Asking and Responding to Questions, Impromptu Speaking, Art of Extempore & Presentations, Role Play, Delivering Welcome Speech, Vote of Thanks, Group Discussion.	**4**
Total		**15**

Lab Session	Activities	Duration (Hrs)
1	Listening 1: Listen to the audio and answer the questions (IELTS)	**2**
2	Listening 2 : Listen to the audio and Summarize (Ted Talks)	**2**
3	Grammar: Correct the sentences and understand the business usages.	**2**
4	Vocabulary: Different ways to improve vocabulary and activities	**2**

5	Writing Skills 1: Formal writing such as Job Application, Leave Application, Enquiry and Complaint Letter.	**2**
6	Writing Skills 2: Different Styles of writing and Paragraph Writing (Descriptive, Technical)	**2**
7	Writing Skills 3: Technical Writing, Report Writing; Progress, Accident Report, Event Report.	**2**
8	Reading Activity 1: Communication Case Studies	**2**
9	Reading Activity 2: IELTS based Comprehension Skills	**2**
10	Reading Activity 3: Research Articles and Technical Documents	**2**
11	Reading Activity 4: Literary Reading and Discussion	**2**
12	Speaking Activity 1: IPA Pronunciation and Phonetics Exercises	**2**
13	Speaking Activity 2: Delivering speeches and Mastering the Art of Public Speaking	**2**
14	Speaking Activity 3: Preparing and Participating Group Discussions / Elevator Speeches	**2**
15	Speaking Activity 4: Oral/PPT Presentation with Q&A Session	**2**
Total		**30**

Text Books:

1. Raymond Murphy, Essential English Grammar in Use, Cambridge University Press; 2015

Reference Books:

1. Michael Swan, Practical English Usage, Oxford, 3rd Edition; 2005
2. David F. Beer, Writing and Speaking in the Technology Professions: A Practical Guide, Wiley-IEEE Press; 2nd Edition, 2003
3. Sunita Mishra, C. Muralikrishna, Communication Skills for Engineers, Pearson Education; 2011
4. Clifford Whitcomb, Leslie E. Whitcomb, Effective Interpersonal and Team Communication Skills for Engineers, Wiley–Blackwell; Nil edition, 2013.
5. Krishnaswami, N and Sriraman, T, Creative English for Communication, Macmillan.Saran Freeman, Written Communication in English, Orient Longman.

E Sources -

1. https://www.google.com/url?q=https://onlinecourses.nptel.ac.in/noc19_hs19/&sa=D&source=editors&ust=1654924489543365&usg=AOvVaw0vWlA1-FXdmtGD4TbPCXo-
2. https://www.google.com/url?q=https://onlinecourses.nptel.ac.in/noc19_hs22/&sa=D&source=editors&ust=1654924489545718&usg=AOvVaw1JiV6Z4RihjTKbm8Sd2HDC
3. https://takeielts.britishcouncil.org/take-ielts/prepare/free-ielts-practice-tests/listening/section-1

Program:	B. Tech. CSE (AI & ML)					Semester: II	
Course:	AEC I-German					Code:	BSH22AE02
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	1	2	-	10	10	30	50

Prior knowledge of
English Language is essential.

Course Objectives:

This course aims at enabling students,

1. To get familiar with the basics of German language and develop their interest in the language.
2. To identify the desired information while reading and listening simple German texts.
3. To acquire basic knowledge of German speaking countries.
4. To frame simple sentences in German.

Course Outcomes:

After learning the course, the students should be able to:

1. Demonstrate an understanding of simple texts in German.
1. Apply basic grammar rules to frame simple sentences in German.
2. Develop simple dialogues in German reflecting situations encountered in daily life.
3. Construct simple texts in German.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	**Introduction to German Language** **Topics:** Greetings; Alphabet; Numbers; Days; Months; Seasons; Personal details; Family; Hobbies; Self-introduction; Things of day-to-day use; Food & Beverages; Buying goods of day-to-day use; Clock time; Basic knowledge of German speaking countries **Listening skills:** Listen and understand spellings, numbers, clock time, details of persons, short easy day-to-day conversations in German **Reading skills:** Read and comprehend from visiting cards, brief profiles, simple instruction boards and advertisements, short easy texts, short messages, short letters, and emails in German, read texts aloud and respond by answering questions accordingly	4
II	**Basic German Grammar and Sentence Structure** ● **Personal Pronouns:** Nominative ● **Verbs and Verb-Conjugation:** Regular, irregular, separable, modal auxiliaries ● **Types of Articles:** Definite, indefinite, negative, possessive ● **Cases:** Nominative, accusative ● **Prepositions:** With accusative case ● **Types of the sentences:** Declarative, interrogative, imperative ● **Tenses:** Present Solving simple grammar exercises to get used to basic sentence structure in German	4

III	**Speaking Skills** ● Spelling and pronouncing words correctly ● Giving brief self-introduction in German ● Asking for personal details and providing the required information ● Requesting for things of day-to-day use and reacting on requests in appropriate manner Writing short easy German dialogues and presenting them with expressions	3
IV	**Writing Skills** ● Writing short easy sentences in German ● Using German punctuation and orthographic rules correctly in given texts ● Taking dictation for words and simple sentences ● Correcting errors in given texts Writing simple texts, short messages, letters emails on given topics	4
	Total	**15**

Lab	Activities	Duration (Hrs)
1	**Vocabulary:** Exercises to recall and enhance vocabulary	2
2	**Listening 1:** Listen to the audio and repeat (phonetics)	2
3	**Listening 2:** Listen to the audio and select the correct option	2
4	**Reading 1:** Read short easy texts and fill up the information in table	2
5	**Reading 2:** Read short easy texts and mark true or false	2
6	**Reading 3:** Read short easy texts and answer the questions	2
7	**Grammar 1:** Solve simple grammar exercises	2
8	**Grammar 2:** Construct correct sentences by applying grammar rules	2
9	**Speaking 1:** Spell and pronounce the words correctly	2
10	**Speaking 2:** Give your short introduction	2
11	**Speaking 3:** Frame simple questions, requests and reply	2
12	**Writing 1:** Fill up simple data in registration forms	2
13	**Writing 2:** Correct errors in given draft	2
14	**Writing 3:** Fill in the sentences and rewrite the texts, short messages, emails, and letters	2
15	**Presentation:** Basic information of India and German speaking countries in German	2
	Total	**30**

Text Books:

1. Menschen A1.1: Sandra Evnas, Angela Pude, Franz Pecht, Hueber Verlag Ismaning Germany, 2016

Reference Books:

1. Netzwerk A1: Dengler, Rusch, Schmitz, Sieber, Ernst Klett Sprachen, Stuttgart Germany, Goyal Publishers & Distributors, Delhi, 2015
2. Linie 1: Kaufmann, Moritz, Rodi, Rohrmann, Sonntag,Klett-Langenscheidt GmbH, München Germany, Goyal Publishers & Distributors, Delhi, 2018
3. Tangram aktuell 1: Dallapiazza, Eduard von Jan, Schönherr, Max Hueber Verlag, Ismaning, Germany, Goyal Publishers & Distributors, Delhi, 2005

e-sources:

1. **NPTEL Course lectures (IIT Madras) link:** https://onlinecourses.nptel.ac.in/noc24_hs29/preview
2. **DW Learn link:** https://learngerman.dw.com/en/learn-german/s-9528
3. **Goethe-Institut Link:** https://www.goethe.de/en/spr/ueb.html
2. **Easy German link:** https://www.easygerman.org

Program:	B. Tech. CSE (AI & ML)					Semester: II	
Course :	AEC-I Japanese					Code:	BSH22AE03
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	1	2	-	10	10	30	50

Prior knowledge of:

1. English/Marathi/Hindi language for learning Japanese language.

Course Objectives:

This course aims at enabling students,

1. To be aware of Japanese Scripts (Hiragana, Katakana) and basic Kanjis
2. To familiarize themselves with the Japanese language and use basic greetings in day-to-day life.
3. To express themselves using basic sentences and develop cross cultural skills and understanding of gestures, family and community, perceptions.
4. To develop language skills namely Listening, Speaking, Reading and Writing skills for socializing, at basic level.

Course Outcomes:

After learning the course, the students should be able to:

1. Recognize Japanese scripts through oral and written communication.
2. Interact with the people using Japanese greetings in to their day-to-day life.
3. Demonstrate the basic Kanjis with meanings.
4. Construct simple demonstrative sentences.

Detailed Syllabus:

Unit	Description	Duration [Hrs]
I	Introduction: **Hiragana Script.** Listening: Short video skit on self-introduction Speaking: Song of greetings. Reading: Hiragana words Writing: Japanese scripts (Hiragana) Test on Hiragana	5
II	Introduction: **Katakana script** Listening: English words Speaking: Song on body parts. Reading: Katakana words Writing: Locating countries on map, Word hunt.	4
III	Introduction to Kanjis Writing: Learn to write kanjis with stroke order.	3
IV	わたし は マイク.ミラー です。 Listening: Conversation based on L-1 Speaking: Self introduction Reading: Lesson reading no.-1 Writing: Writing about yourself. Grammar: Introduction to1.Particles (は、か、も、か) 2.Verbs (です、ではありません)	3
	Total	**15**

Practical/Lab sessions		
Lab session	**Activities**	**Duration (Hrs)**
1	Writing Skill 1: Hiragana script	2
2	Speaking skill 1: Japanese greetings	2
3	Reading Skill 1: Reading and recognizing 'Hiragana' words	2
4	Listening Skill 1: Listening and writing 'Hiragana' words	2
5	Reading Skill 2: Reading 'Katakana' words	2
6	Writing Skill 2: World map activity	2
7	Speaking Skill 2: Self introduction	2
8	Listening Skill 3: Listening and identifying the numbers.	2
9	Reading Skill 3: Reading Numbers with writing practice.	2
10	Speaking Skill 3: Practicing Japanese Greetings.	2
11	Writing Skill 3: Creating Kanjis chart using strokes, 'Kun Yomi' and 'On Yomi'	2
12	Listening Skill 4: Conversation in the office	2
13	Speaking Skill 4: Dialogues between people of different nationality.	2
14	Reading Skill 4: Chapter-1 reading	2
15	Writing Skill 4: Basic sentence formation using grammar.	2
	Total	**30 Hrs.**

Textbook:

1. Minna no Nihongo Part Iand II Publication :GOYAL PUBLISHERS & DISTRIBUTORS PVT. LTD. ,Author: TsuruoYoshiko (Compiled),Edition:2018
2. Nihongo Shoho Publication:JALTAP, Author:JALTAP(With permission of Japan Foundation, Tokyo), Edition:April 2008

Reference Books:

1. Genki
2. MOMO Author: Japan Foundation, NewDelhi, Publication: Goyal Publisher & Distributors(P) Ltd.,Edition:October2007
3. MOMO Japanese workbook Japan Foundation, New Delhi, Publication: Goyal Publisher & Distributors(P)Ltd.,Edition:October2007
4. MOMO Japanese workbook Japan Foundation, New Delhi,Publication:Goyal Publisher & Distributors(P)Ltd.,Edition:October2007

E references:

1. https://onlinecourses.nptel.ac.in/noc23_hs120/preview
2. https://nptel.ac.in/courses/121104005

Program:	B. Tech. CSE (AI & ML)					Semester: II	
Course:	AEC I-Business Story telling					Code:	BSH22AE04
Credit	Teaching Scheme (Hrs./Week)			Evaluation Scheme			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	1	2	-	10	10	30	50

Prior Knowledge: Basic knowledge of English Language is essential.

Course Objectives: This course aims at enabling students,

1. To understand storytelling as one of the tools of influential communication.
2. To strengthen their creativity, critical thinking and social skills.
3. To use stories to face leadership, management and professional challenges.

Course Outcomes: After learning the course, the students will be able to

1. Identify nuances of storytelling method as an influential communication
2. Demonstrate the ability to engage and inspire others through the development of narratives, tone and style
3. Apply storytelling techniques to communicate effectively in a business context
4. Develop stories to build, maintain professional relationships, deliver messages and motivate others toward action.

Detailed Syllabus:

Unit	Description	Duration (Hrs)
I	**Concept and Scope:** What is a story? A Brief History & Importance of Storytelling, Basics of Storytelling - Entertainment, Engagement, Personalization, Critical Thinking, Observation Skills in Storytelling, Benefits of Storytelling, Storytelling in Engineering, Business Storytelling, Activity: Analysis of Steve Jobs Commencement Speech at Stanford (2005)	3
II	**Process of Storytelling:** Elements of a Story - Context and Relevance, Style and Detailing, Plot, and Characters, The Flow of the Story - Relevance - Action - Result, Know the Purpose - Inspire Action, Educate People, Showcase Values, Build Collaboration, Know your Audience - Educational, Social Background and Age, Developing Narratives: Characteristics of a Narrative, Data Visualization, Presenting a Word Picture, Triggering Emotions of the Audience, Choosing Media - Audio, Written, Oral and Digital Storytelling **Activity:** Analysis of a Short Story: **'The Three Hermits by Leo Tolstoy', The Last Painting by O' Henry**	4
III	**Types of Stories** - Customer Story, Origin Story, Event Story, Product Stories, Storytelling Techniques for Presentations, Using Power Words Effectively, Using Narratives to Manage Conflicts, Using a Narrative to Interpret the Past and Shape the Future, Storytelling in Marketing, Story Strategies - Using Anchor Stories **Case studies** - Brand storytelling -Steve Jobs / Jack Maa - Product Presentation, Lido Anthony "Lee" Iacocca.	4

IV	**Crafting a Story** Crafting a Story from a Picture/an Idea/Situation/Artifacts, Storyline - Beginning / Motive / Struggle / Achievement, Six-word Story - Memoirs to Being with, Detailing of Character and the Context, Delivering a Story – Tone / Emotions / Voice Modulation **Activity**-Developing and Delivering Presentation through Storytelling on the Given Situation/Context	4
Total		**15**
Lab Session	**Activities**	**Duration (Hrs)**
1	Basic of Storytelling: Using Five Senses in storytelling activity and Elements of Storytelling	2
2	Analysis of a Short Story: 'The Three Hermits by Leo Tolstoy', and The Last Painting by O' Henry.	2
3	Character Study: Create a detailed character profile of a fictional character, including their background, motivations, and personality traits. Write a short story or scene that showcases this character in action	2
4	Personal Storytelling: Write and present a short personal story that highlights a challenge you've faced and how you overcame it	2
5	Collaborative Storytelling: Partner with another student to create a collaborative story. Take turns writing alternating sections, focusing on maintaining a consistent tone and narrative flow.	2
6	Historical Business Story: Research and narrate a significant historical event or moment in a well-known business's journey, focusing on how storytelling played a role in shaping public perception	2
7	Social Impact Story: Develop a story that demonstrates how a business initiative or project positively impacted a community or addressed a social issue	2
8	Customer Success Story: Craft a narrative that showcases a customer's journey with your fictional business	2
9	Change Management Story: Design a narrative that communicates a change initiative within a company, addressing challenges, resistance, and the ultimate benefits of the change	2
10	Investor Pitch Story: Craft a persuasive story for a startup pitch. Highlight the problem, solution, market opportunity, and potential for growth in a captivating way	2
11	Leadership Story: Compose a story that illustrates effective leadership qualities and strategies. Highlight a leader's ability to motivate, inspire, and guide a team toward Success	2
12	Cultural Storytelling: Explore how storytelling can bridge cultural gaps in a global business context. Share a story that demonstrates cultural sensitivity and Understanding	2
13	Ethical Dilemma Story: Present a complex ethical dilemma faced by a business or individual. Use storytelling to explore various perspectives and potential solutions	2

14	Marketing Campaign Story: Design a storytelling-based marketing campaign for a specific product or service launch, incorporating different media and channels	2
15	Crisis Turnaround Story: Narrate a scenario where a business successfully navigated a crisis through strategic communication and storytelling, ultimately regaining trust and reputation.	2
Total		**30**

Text Books:

1. Kendall Haven, Story Smart, Libraries Unlimited, 2014

Reference Books:

1. Kendall Haven, Story Proof, Libraries Unlimited, 2007.
2. Rob Biesenbach, Unleash the Power of Storytelling: Win Hearts, Change Minds, Get Results, Eastlawn Media, 2018.
3. Yiannis Gabriel, Storytelling in Organizations: Facts, Fictions, and Fantasies, Oxford University Press, 2011.

E-resources:

1. The Art of Business Storytelling | AmeenHaque | Talks at Google , https://www.youtube.com/watch?v=77FUr6ZsWjY
2. Marketing Storytelling - https://www.referralcandy.com/blog/storytelling-examples/
3. 5 examples of great storytelling from Jack Ma https://www.youtube.com/watch?v=3nHOxONWfEs
4. Six words story - Nicole Kahnhttps://www.youtube.com/watch?v=16sY1iLc2d4
5. Kevin Hart - Telling great stories https://www.youtube.com/watch?v=vn_L4OPU_rg

Program:	B. Tech. CSE (AI & ML)					Semester: II	
Course:	Life Skills II					Code:	BSH22CC02
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	TW	OR	PR	Total
2	-	4	-	100	-	-	100

Prior knowledge of: Nil

Course Objectives: This course aims at:

1. To equip them with essential skills and knowledge that complement their academic education, preparing them to excel not only as engineers but also as well-balanced individuals.
2. To develop students' vital life skills that promotes personal growth, resilience, and success in their academic journey and beyond.

Course Outcomes:

After learning the course, the students will be able to:

1. Demonstrate the ways to nurture their passion.
2. Develop skills growth mindset to be successful in personal and professional life.
3. Demonstrate adaptability and flexibility for any environment.
4. Apply essential skills for successful and happy life management.

Course Guidelines:

1. The students are instructed to perform all the activities suggested by the course faculty.
2. The assessment of life skills activities will be done on the basis of students' performance, attitude, behavior and understanding of subject.
3. The students are suggested to attend all the sessions of the life skills course.

Detailed Syllabus

Unit	Description	Duration (Hrs)
I	**Nurture Your Passion** Developing Hobbies- Importance, Ways and Benefits , Exploring Skills - Singing/Painting/Dancing etc, Sports: Basketball, Table tennis, Football and Volleyball, Performing Arts: Painting/ Sketching/ Drawing, Stage performance, Let's Play to Learn - games and play forms possible, like, Puzzles & Brainteasers, quiz. ● Activity/Worksheet 1: Nurture Your Passion ● Activity/Worksheet 2: Importance of Hobbies ● Activity/Worksheet 3: Stage Performance ● Activity/Worksheet 4: Performing Arts: Painting/ Sketching/ Drawing	15
II	**Lead Yourself - Growth Mindset** Understanding the concept for personal development, Embracing change: Coping with the dynamic nature of life, Resilience and perseverance: Overcoming obstacles and setbacks, Developing self-leadership skills and taking initiative/ responsibilities. ● Activity/Worksheet 5: Understanding the Concept of Personal Development ● Activity/Worksheet 6: Embracing Change: Coping with Dynamic Nature of Life ● Activity/Worksheet 7: Resilience & Perseverance: Overcoming Obstacles & Setbacks	15

III	**Adaptability and Flexibility** Adaptability in a rapidly changing world, Problem-solving and decision-making in dynamic situations. Approaching Problem Differently, Embracing uncertainty: Coping with ambiguity and making the most of new opportunities, Flexibility in teamwork: Navigating diverse team dynamics effectively ● Activity/Worksheet 8: Adaptability in Rapidly Changing World ● Activity/Worksheet 9: Flexibility in Teamwork	15
IV	**Life Management** Financial Literacy-Saving is earning, Value of money, Coping up with Virtual Life and Reality, Understanding the responsibilities and impact of Global Citizenship, Environmental awareness and sustainable practices (v) Social responsibility: Contributing positively to the community. ● Activity/Worksheet 10: Financial Literacy & Virtual Life	15
	Total	**60**

Reference Books:

1. "Mindset: The New Psychology of Success" by Carol S. Dweck Publisher: Ballantine Books
2. "The Financial Diet: A Total Beginner's Guide to Getting Good with Money" by Chelsea Fagan and Lauren VerHage
3. "Grit: The Power of Passion and Perseverance" by Angela Duckworth Publisher: Scribner, 2018

E Sources -

1. Skills You Need (www.skillsyouneed.com): This website offers comprehensive information and practical guidance on a wide range of life skills, including communication, time management, problem-solving, and more
2. Mind Tools (www.mindtools.com): Mind Tools provides resources on personal effectiveness, leadership, communication skills, and other essential life skills to enhance professional and personal development
3. TED Talks (www.ted.com): TED Talks offer inspiring and informative speeches by experts and thought leaders covering various life skills topics, including resilience, emotional intelligence, and personal growth
4. Very well Mind (www.verywellmind.com): This website covers mental health, emotional well-being, and self improvement topics that contribute to overall life skills development

Vision and Mission of Applied Sciences and Humanities (AS & H) Department

Vision

To provide value-added quality education that promotes essential technical skills, critical-thinking, communication skills and human values to make impactful contributions to the society.

Mission

Being a student-centric department, our mission is –

1. To develop a strong base of engineering sciences through innovative and experiential learning.
2. To provide excellent harmony of conducive environment and moral support for joyful learning.
3. To strive for overall development of students by providing the right platform to nurture all personality traits.
4. To create research attitude and endeavor innovation, creativity.

Vision and Mission of Computer Sciences & Engineering (AI & ML) Department

Vision

To provide value-based technical Education in Computer Science and Engineering with specialization in Artificial Intelligence and Machine Learning

Mission

M1: To develop technically competent and innovative computer science engineers with in-depth knowledge of Artificial Intelligence and Machine Learning.

M2: To build ethically responsible, knowledgeable and skilled engineers to serve the needs of industry and society at large.

M3: To provide conducive environment and opportunities for holistic development of students